

Nibrexo OS AI

08 — AI Agent Architecture & Automation Engine

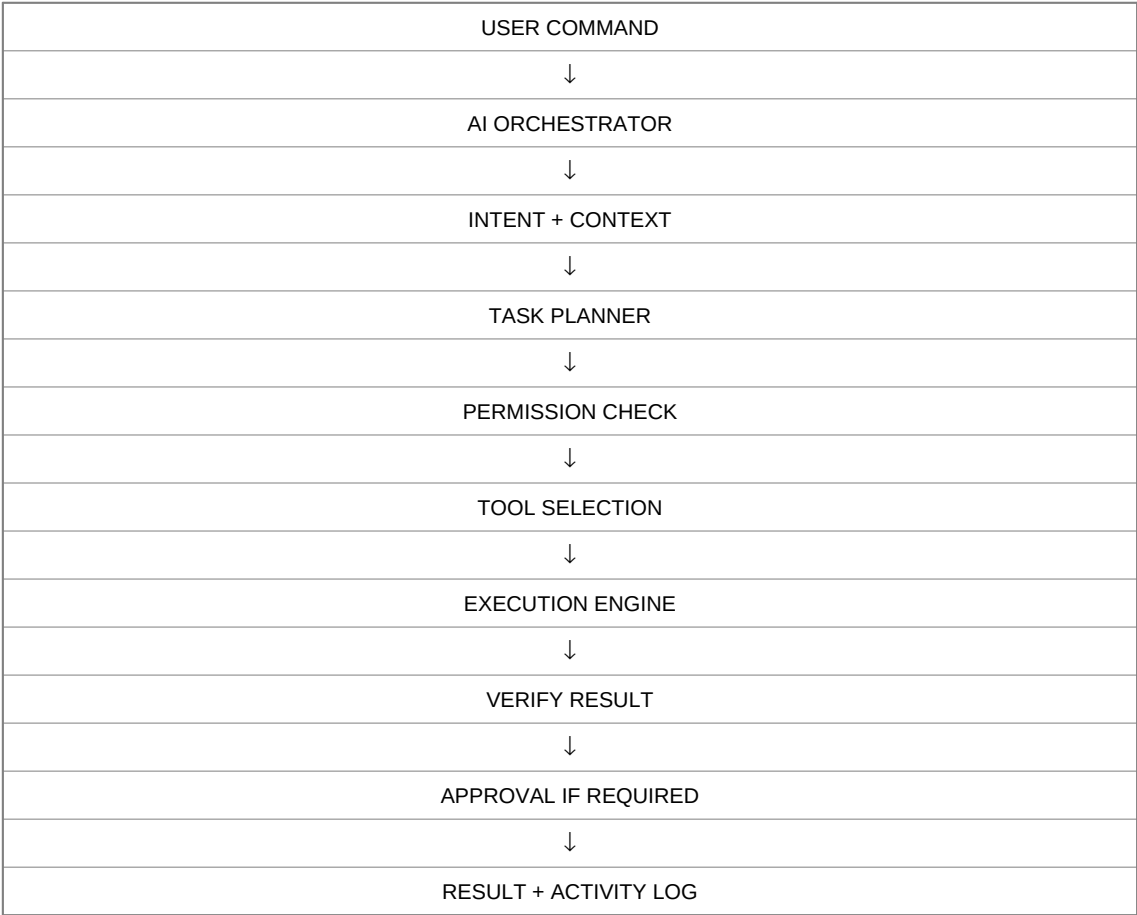
v1.0

Production blueprint for the Nibrexo AI agent brain, tools, memory, permissions, planning and automation engine.

1. Purpose

Nibrexo OS AI ka agent ek controlled business operator hoga jo user ke natural-language commands ko samajh kar approved tools ke through social, content, inbox, CRM aur email workflows execute karega.

2. Agent High-Level Architecture



3. Core Agent Components

Component	Responsibility
Orchestrator	Controls complete agent lifecycle
Intent Analyzer	Understands what user wants
Planner	Breaks complex requests into steps
Permission Guard	Checks whether action is allowed
Tool Registry	Lists available agent tools
Execution Engine	Runs approved tools
Memory Layer	Stores approved useful context

Component	Responsibility
Verifier	Checks action/result
Approval Manager	Stops risky external actions until approval
Audit Logger	Records important agent activity

4. Agent Request Lifecycle



5. Tool Architecture

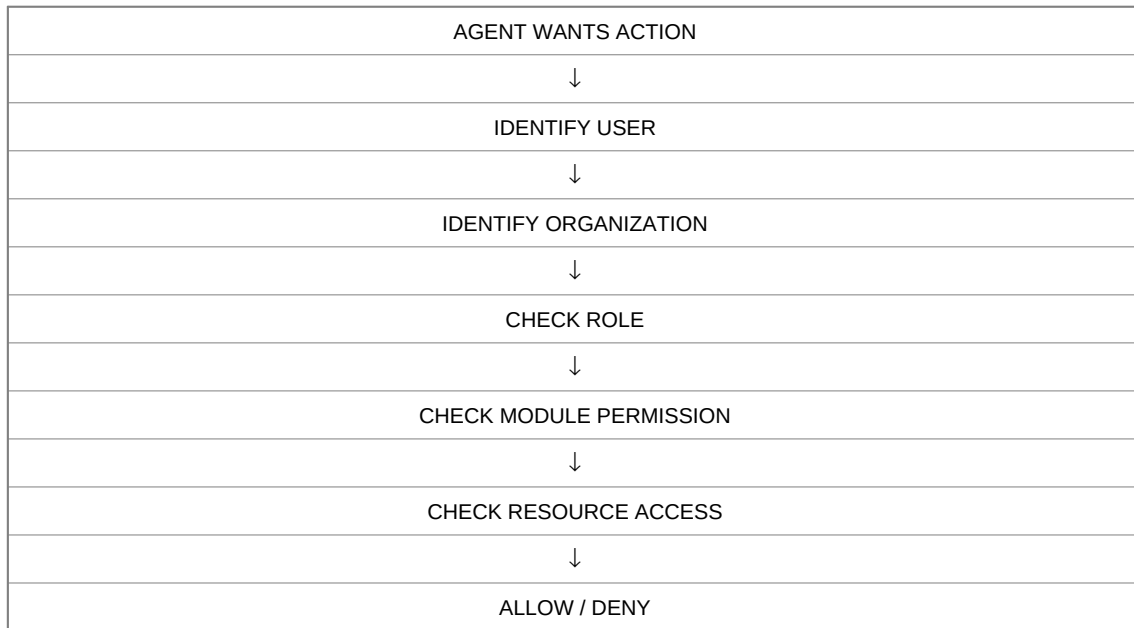
- Every agent capability is exposed as a controlled tool.
- Tools have a defined name, purpose, input schema, output schema and permission requirement.
- Agent cannot invent tools or call arbitrary external services.
- Tool execution happens server-side.
- Failed tools return structured errors.
- Every important tool execution can be logged.

6. Initial Tool Groups

Tool Group	Examples
Social	connect_account, get_accounts, publish_post, schedule_post
Content	create_content, update_content, upload_media, generate_caption

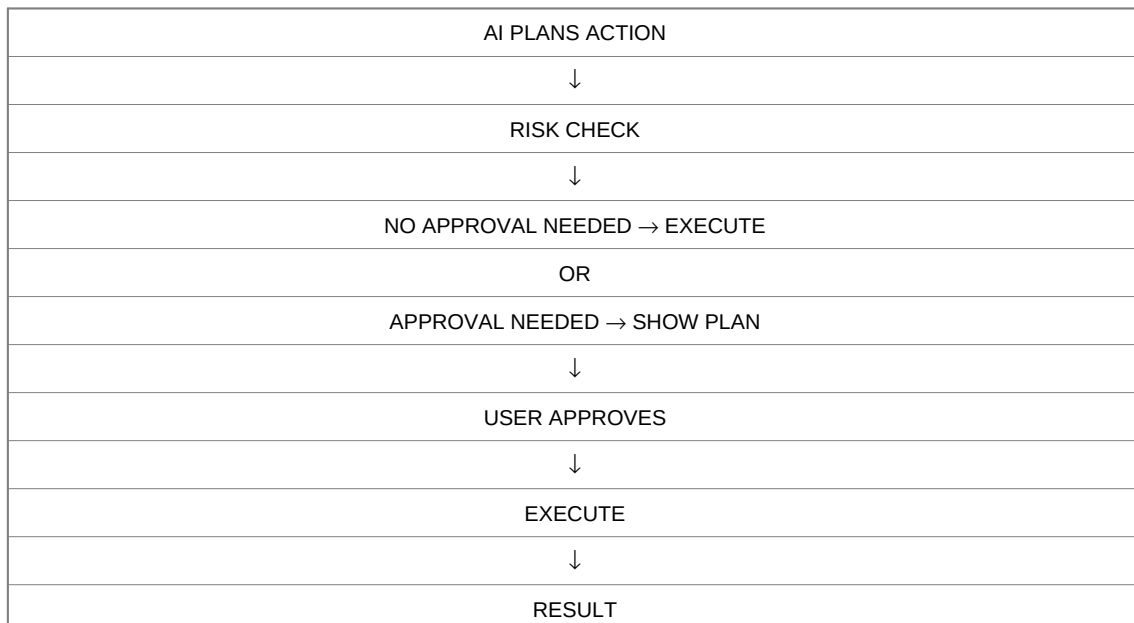
Tool Group	Examples
Inbox	get_conversations, get_messages, draft_reply, send_reply
CRM	create_client, update_client, get_client, add_note
Email	create_template, send_email, create_sequence, pause_sequence
Analytics	get_post_metrics, get_email_status
System	get_settings, create_notification, log_activity

7. Permission Guard



- Permission checks happen before tool execution.
- Backend permission checks are authoritative.
- Frontend visibility is not a security boundary.
- AI cannot grant itself permission.

8. Approval Engine

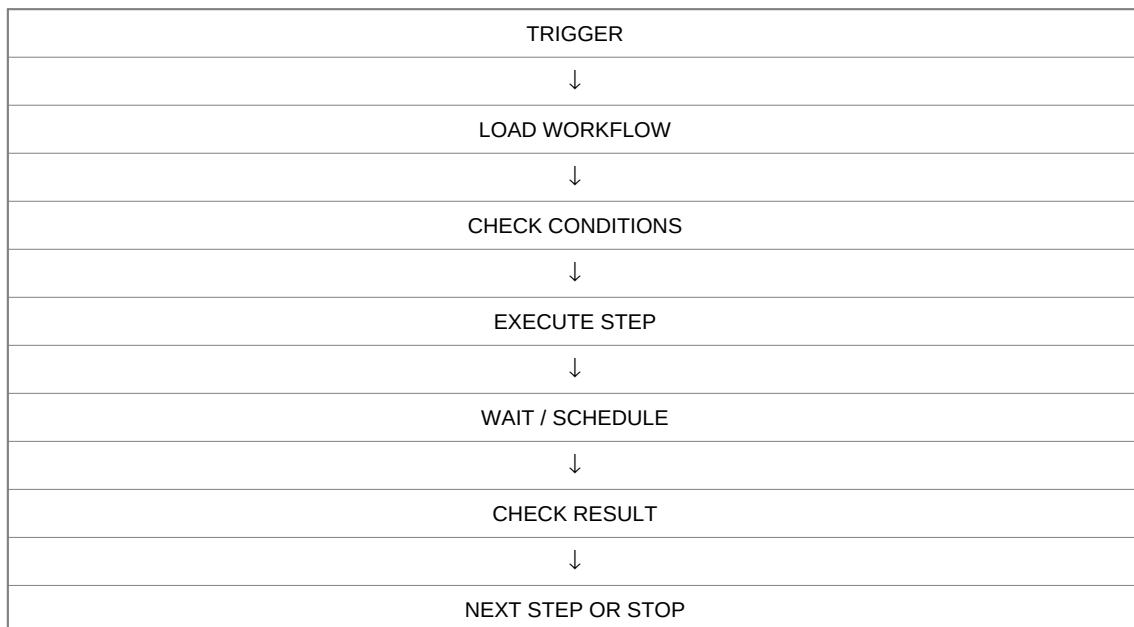


- Publishing, sending external communication or destructive actions can require approval.
- Approval must be explicit.
- Rejected actions are not executed.
- Expired approvals cannot be reused.

9. Memory Architecture

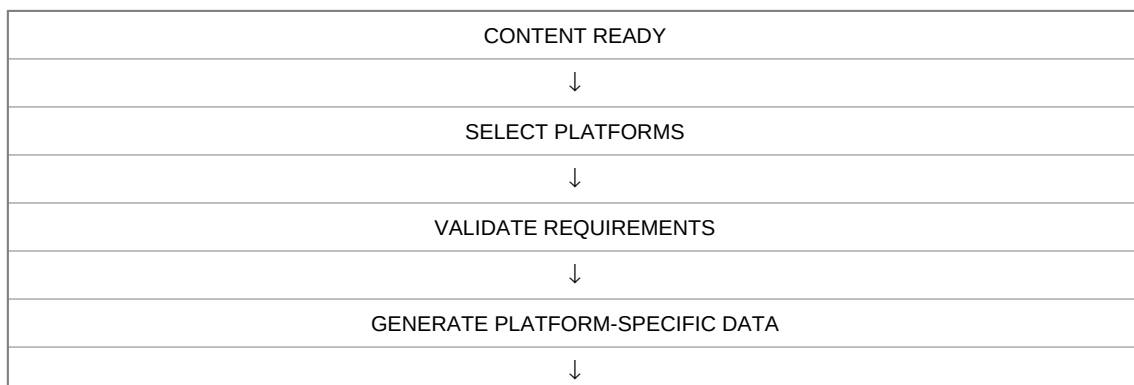
- Short-term context: current conversation/task.
- User preferences: approved working preferences.
- Client context: approved client information.
- Task history: previous execution results.
- Memory must be permission-scoped.
- Sensitive data should not be stored as unrestricted AI memory.

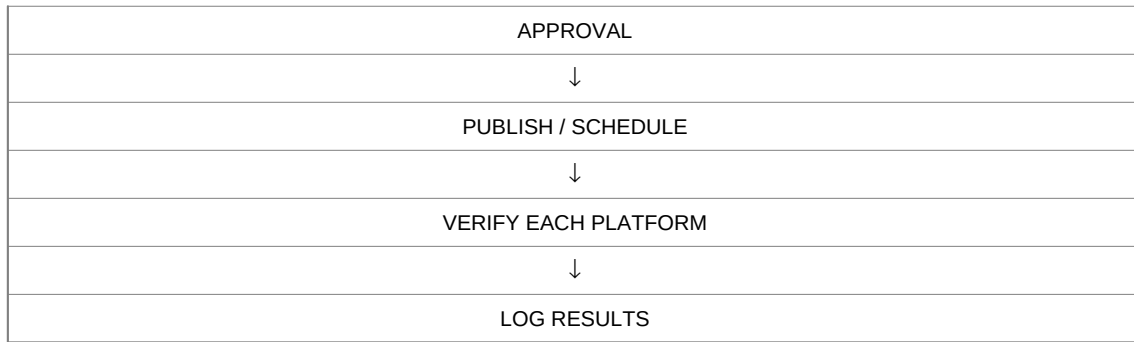
10. Automation Engine



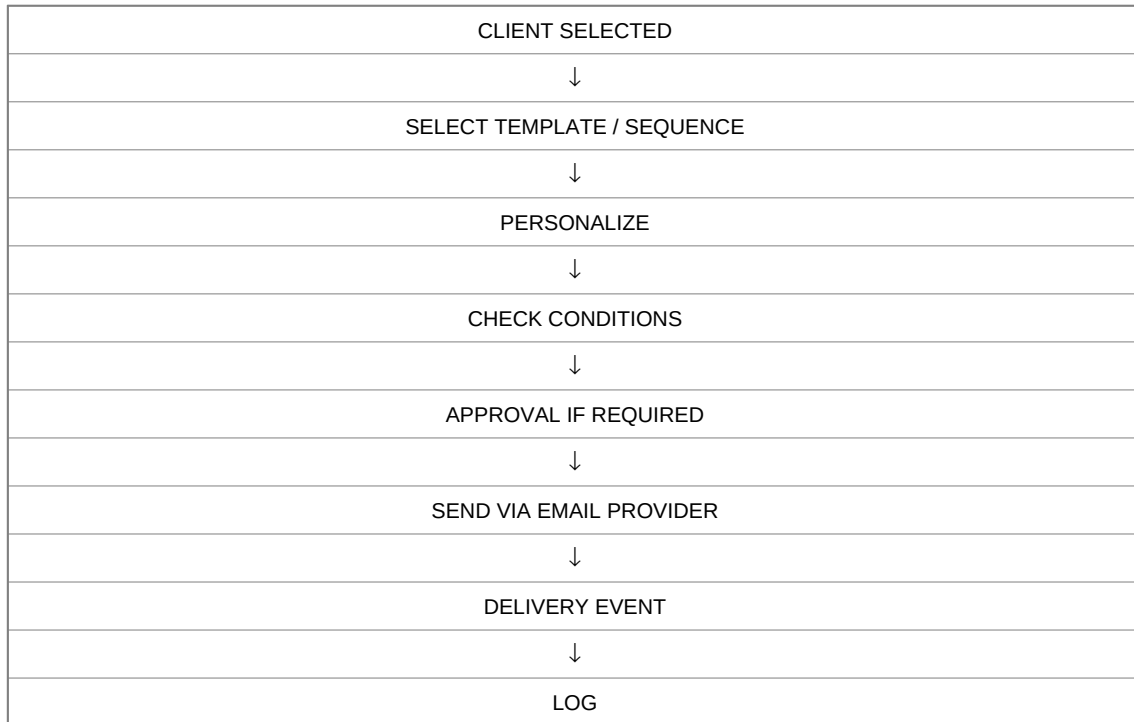
- Triggers can include schedule, user action and supported external events.
- Every automation step has a clear status.
- Failures must stop, retry or continue according to configured policy.
- Duplicate execution protection is required.

11. Social Automation Flow





12. Email Automation Flow



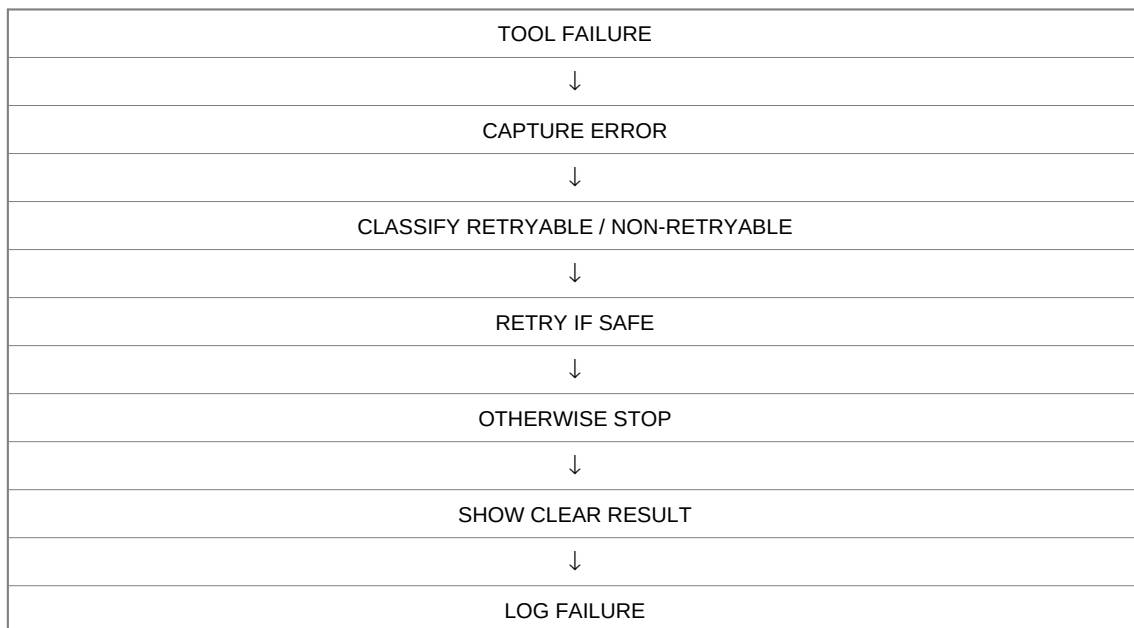
13. AI Reply Flow



14. Agent Safety Rules

- Never bypass authentication or authorization.
- Never expose secrets or API credentials.
- Never invent a successful result.
- Never claim an external action happened until verified.
- Do not send external communication when required approval is missing.
- Do not delete or modify important data without required permission.
- Do not execute an unknown tool.
- Keep an activity trail for significant actions.

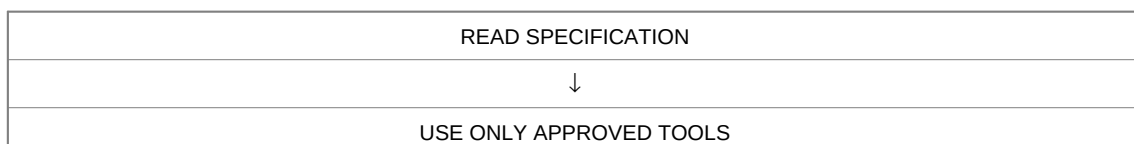
15. Failure & Recovery

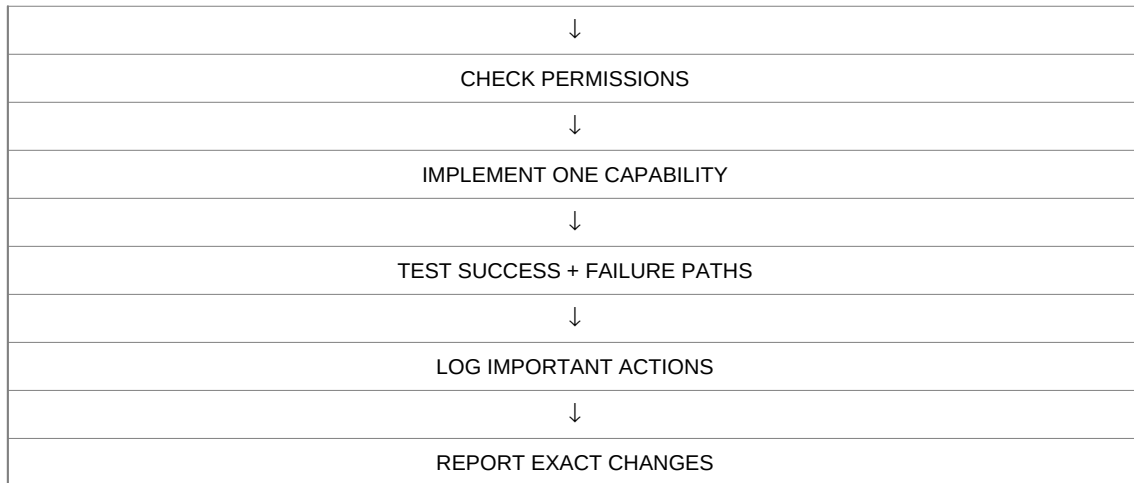


16. Agent State Machine

State	Meaning
RECEIVED	User request received
PLANNING	Agent creating plan
WAITING_APPROVAL	External/risky action requires approval
EXECUTING	Tool is running
VERIFYING	Result is being checked
COMPLETED	Task succeeded
FAILED	Task failed
CANCELLED	User/system cancelled task

17. Agent Development Contract for LLM Arena





The coding agent must not silently create autonomous behavior outside the approved tool registry, permission model or automation rules.

18. Production Readiness Checklist

- Tool schemas documented.
- Permission requirements documented.
- Approval rules implemented.
- Agent states persisted where required.
- Retries are safe and idempotent.
- External action results verified.
- Audit logging enabled.
- Secrets protected.
- Failure paths tested.
- Human override/cancel available.

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